

Which Feature Network Makes a Good Perceptual Loss for Low-Dose CT?

Progress report, Weeks 1 and 2

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Results freeze Oct 18 · first draft Oct 23 · deadline Oct 26

$$\mathcal{L}_\phi = \|\hat{X} - x\|_1 + \lambda_\phi d_\phi(\text{crop}(\hat{x}), \text{crop}(x))$$

- Data: I2I 2025v1, 512×512. Input 100k dose, target normal dose.
- Split: 899 / 182 / 220 slices, patient-level.
- Backbone: lab U-Net. AdamW 1e-4, batch 8, 30 epochs.
- Checkpoint: lowest validation L1, for every loss.
- λ : perceptual term fixed at 95% of the initial loss.
- Metrics: PSNR, SSIM, LPIPS, DISTS, high-band NPS ratio R_H.
- Seeds: 42, 123, 456. Seed 42 is complete.

Control

Only the feature network and the matching rule vary between runs.

ID	Feature network	Family	Matching
C0	none (L1)	reference	-
C1	VGG-19 relu3_2	ImageNet CNN	pointwise
C2	LPIPS-VGG	calibrated IQA	pointwise
C3	DISTS	texture-statistics IQA	statistics
C4	RadImageNet ResNet-50	medical CNN	pointwise
C5	DINOv2-S	self-supervised ViT	pointwise
C6	DINOv2-B	self-supervised ViT	pointwise
C7	MedSigLIP-448	medical VLM	pointwise
C8	DINOv2-S	self-supervised ViT	statistics
C9	CT dose network	task-trained, CT	pointwise
C10	CT dose network	task-trained, CT	statistics

Planned contrasts

Matching: C5 (DINOv2-S, pointwise) vs C8 (DINOv2-S, statistics); C9 (CT dose network, pointwise) vs C10 (CT dose network, statistics).

Size: C5 (DINOv2-S, pointwise) vs C6 (DINOv2-B, pointwise).

Screen only

DINOv2-L, CLIP ViT-B/16.

Pointwise

$$d_{\phi}^{pt} = \sum_I \frac{1}{H_I W_I} \sum_{h, w} \|\bar{\phi}_I(\hat{x})_{hw} - \bar{\phi}_I(x)_{hw}\|_2^2$$

Position-wise match of channel-normalised features.

Local statistics (proposed)

$$d_{\phi}^{st} = \sum_I \frac{1}{C_I |\Omega_I|} \sum_{c, \omega} \left[1 - \frac{2\hat{\mu}\mu + c_1}{\hat{\mu}^2 + \mu^2 + c_1} \cdot \frac{2\hat{\sigma}\sigma + c_2}{\hat{\sigma}^2 + \sigma^2 + c_2} \right]$$

Mean and spread per channel, over 4×4 windows. No cross-correlation term.

Hypothesis

Statistics matching should preserve texture.

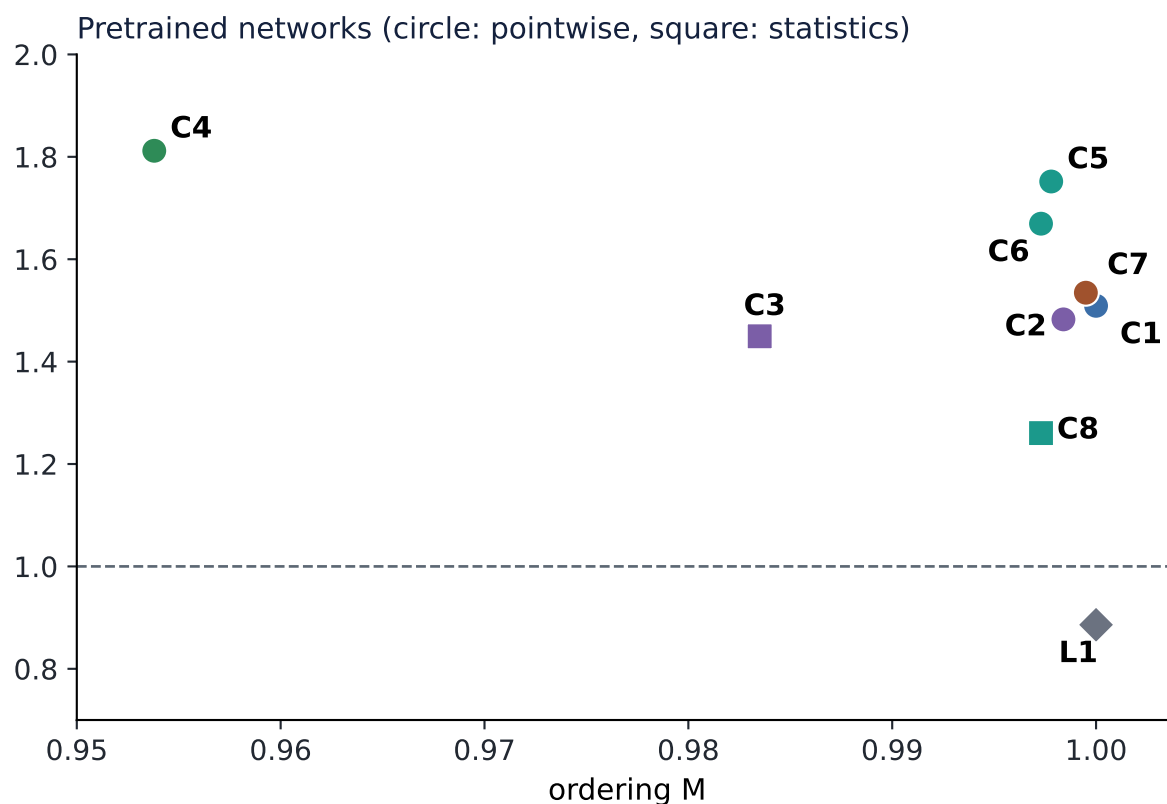
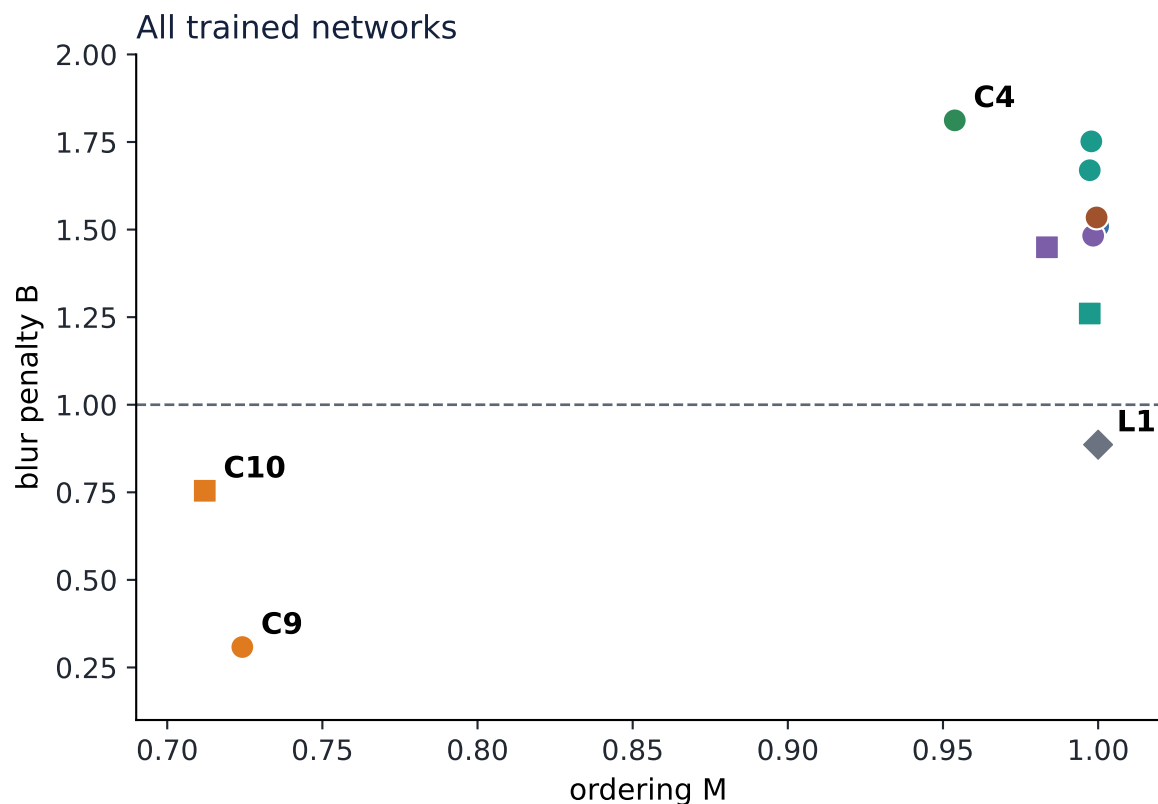
Check	Expected	Observed
Patients train / val / test	disjoint	115 / 34 / 31
Shared patients	0	0
Slices train / val / test	899 / 182 / 220	899 / 182 / 220
Value range	[0, 1]	[0, 1]
Identity test (PSNR, SSIM, LPIPS, R_H)	inf, 1, 0, 1	inf, 1, 0, 1
SSIM, 100k input vs target	< trained outputs	0.742

Result

Data, split and metrics verified.

SSIM

Input 0.742, outputs 0.80 to 0.87. Earlier #261 values (0.49 to 0.68) are most likely a pipeline error.



M: Spearman correlation of distance with dose. B: distance of an MSE-matched blur relative to the 100k image.

- Pretrained networks: $M \geq 0.95$ and B between 1.25 and 1.81.
- Statistics matching lowers B (DINOv2-S: 1.26 against 1.75).
- CT dose network: $M = 0.72$ and $B = 0.31$ (pointwise).

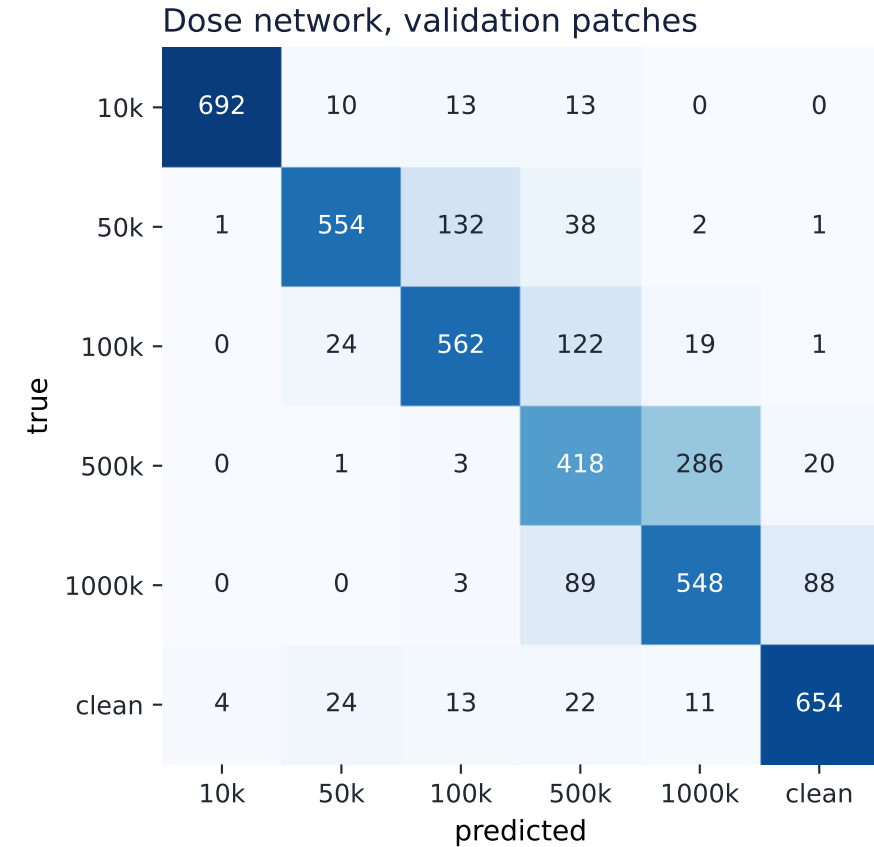
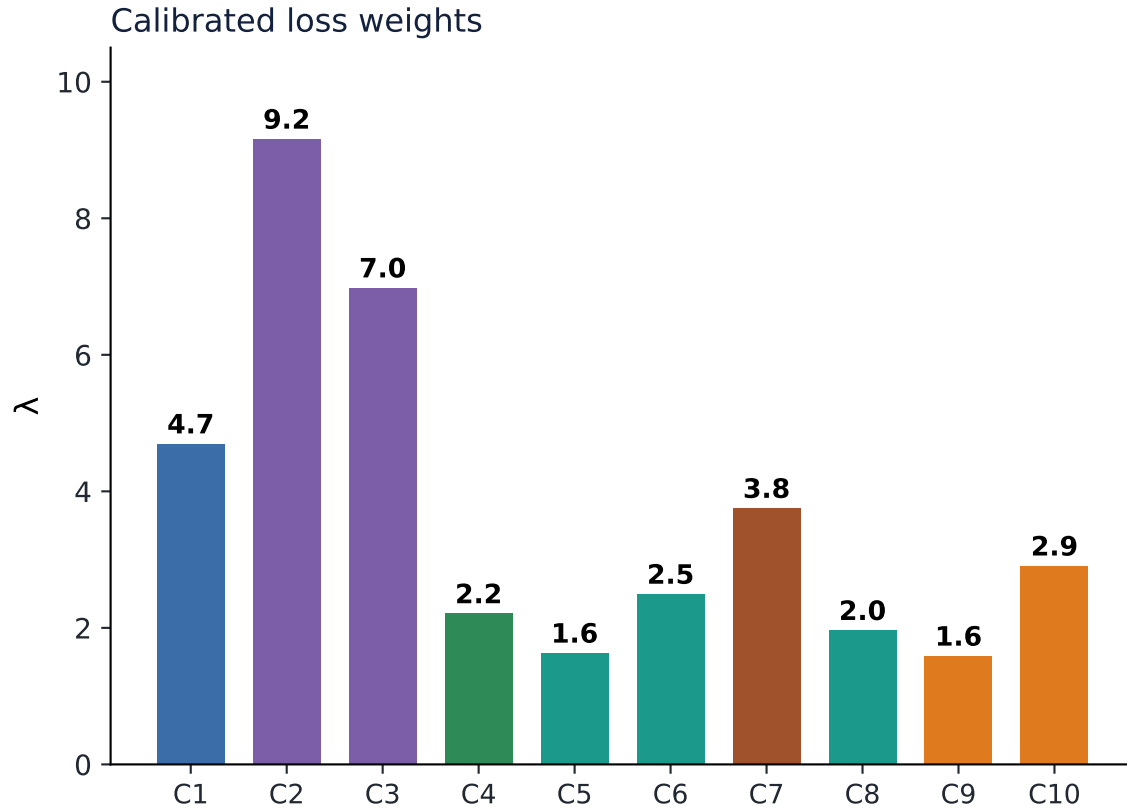
Week 1: screen, blur penalty and dose ratio

week 1

Network	Matching	Blur penalty B	d10k / d100k
Pixel L1	-	0.886	2.46
VGG-19 relu3_2	pointwise	1.509	1.50
VGG-19 relu3_2	statistics	1.418	1.26
LPIPS-VGG	pointwise	1.482	1.43
DISTS	statistics	1.449	1.62
RadImageNet ResNet-50	pointwise	1.812	1.85
RadImageNet ResNet-50	statistics	1.396	1.55
DINOv2-S	pointwise	1.752	3.41
DINOv2-S	statistics	1.260	1.58
DINOv2-B	pointwise	1.669	2.90
DINOv2-B	statistics	1.250	1.56
DINOv2-L	pointwise	1.874	2.84
DINOv2-L	statistics	1.278	1.54
CLIP ViT-B/16	pointwise	1.571	1.58
CLIP ViT-B/16	statistics	1.419	1.29
MedSigLIP-448	pointwise	1.535	1.75
MedSigLIP-448	statistics	1.537	1.47
CT dose network	pointwise	0.308	1.28
CT dose network	statistics	0.754	1.22

Shaded rows: network and matching not used in C0 to C10 (screen only or untrained matching).

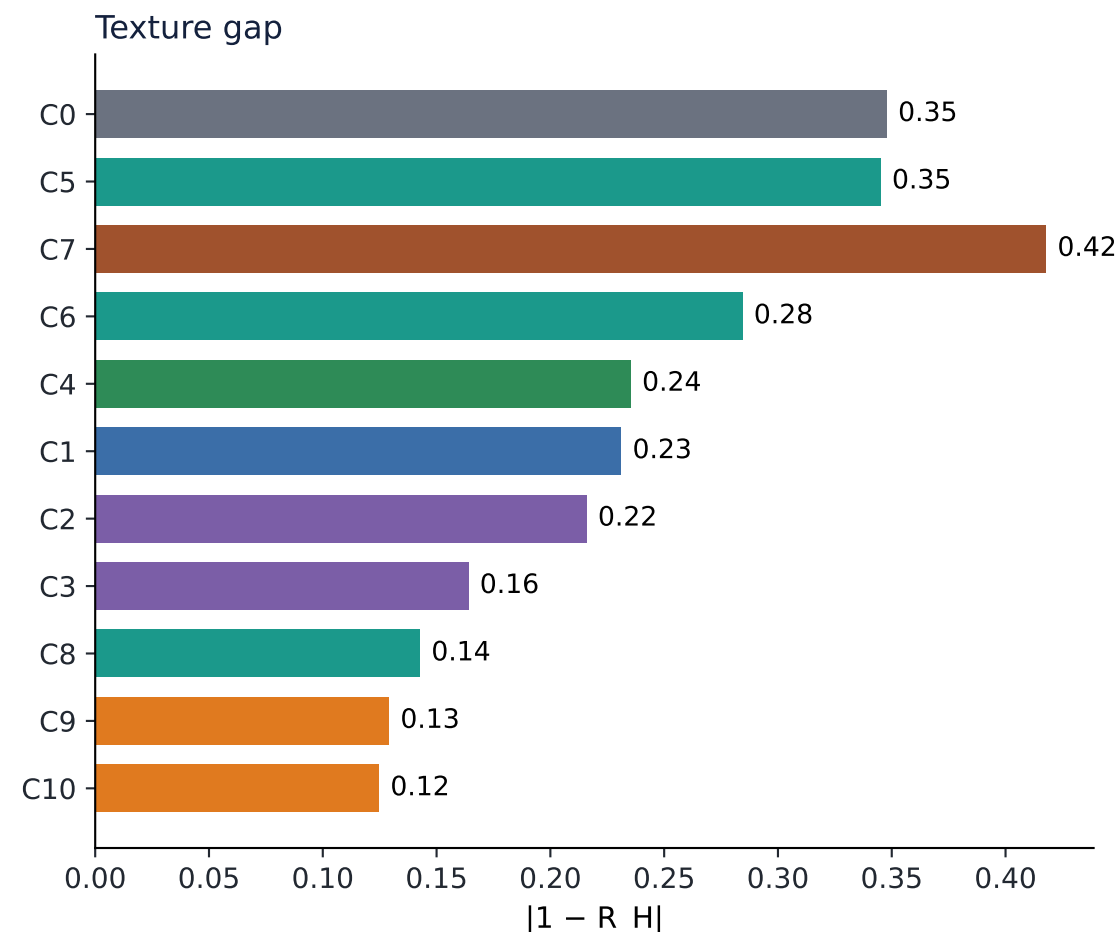
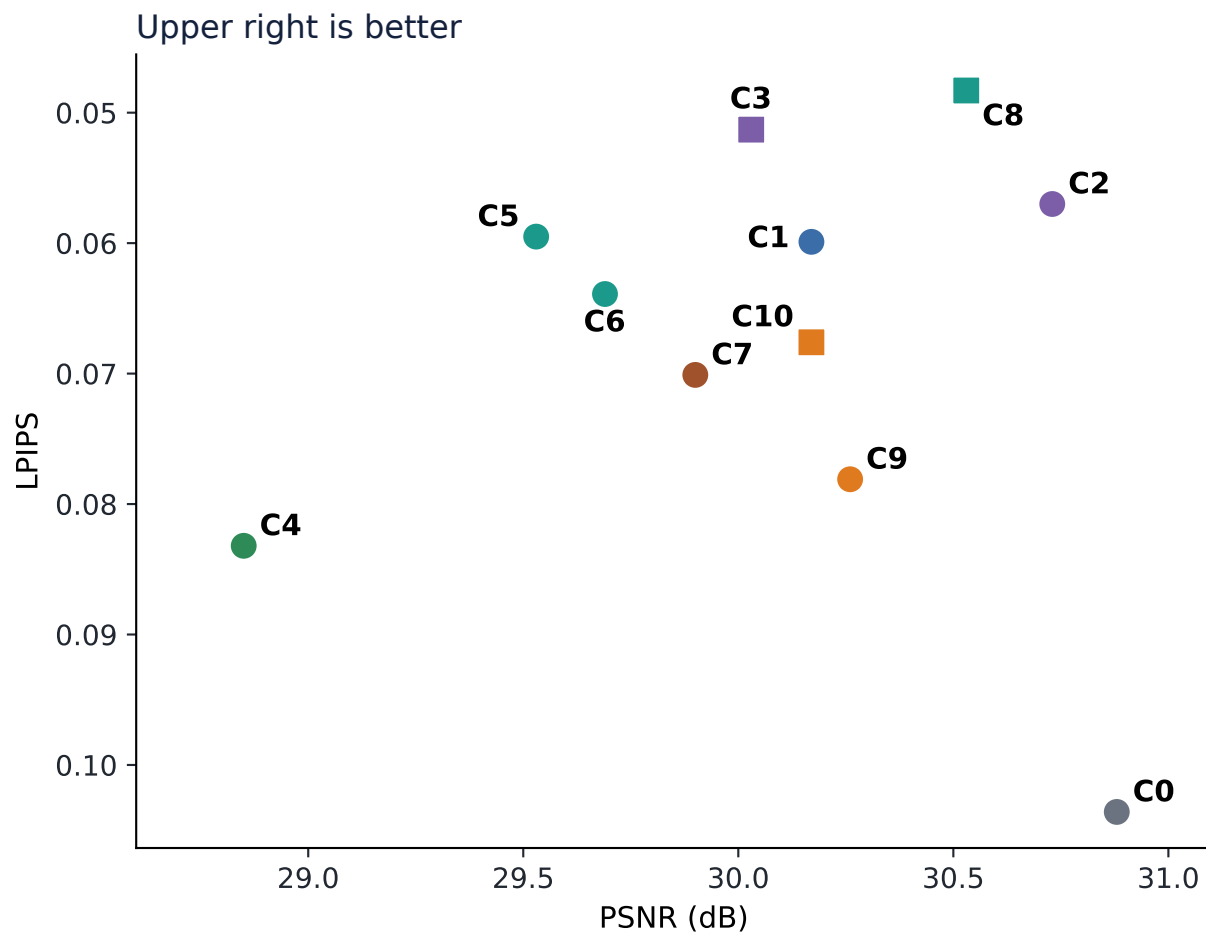
Green: best. Red: worst (higher is better; pixel L1 excluded). d10k / d100k: mean distance at 10k relative to 100k.



- Calibration: $S1 = 0.274$ for all rows. λ ranges from 1.6 to 9.2. No divergence.
- Dose network: 128×128 patches, 21.6k train, 10 epochs, 6.7 min. Validation accuracy 78.5%.
- Errors concentrate at 500k / 1000k and 50k / 100k.

Configuration	PSNR ↑	SSIM ↑	LPIPS ↓	DISTS ↓	R_H → 1	ms/step
LDCT input (100k)	23.04	0.742	0.284	0.164	6.17	-
C0 L1	30.88	0.872	0.104	0.112	0.652	118
C1 VGG-19	30.17	0.862	0.060	0.069	0.769	129
C2 LPIPS-VGG	30.73	0.864	0.057	0.063	0.784	141
C3 DISTS (circular)	30.03	0.805	0.051	0.065	0.836	147
C4 RadImageNet	28.85	0.834	0.083	0.084	0.765	134
C5 DINOv2-S pointwise	29.53	0.865	0.060	0.067	0.655	150
C6 DINOv2-B pointwise	29.69	0.857	0.064	0.075	0.716	201
C7 MedSigLIP	29.90	0.858	0.070	0.071	0.582	1485
C8 DINOv2-S statistics	30.53	0.866	0.048	0.060	0.857	153
C9 CT net pointwise	30.26	0.846	0.078	0.100	0.871	128
C10 CT net statistics	30.17	0.854	0.068	0.093	0.875	132

- C8 (DINOv2-S, statistics) is the best trade-off: −0.35 dB for the best LPIPS and DISTS.
- C5 (DINOv2-S, pointwise) and C7 (MedSigLIP, pointwise) remain smooth ($R_H \leq 0.66$). C4 (RadImageNet, pointwise) loses 2 dB. Spreads over slices; one seed.



Circle: pointwise. Square: statistics. Colour: network family.

Pair	Δ PSNR	Δ SSIM	Δ LPIPS	Δ DISTS	R_H
C5 → C8 (DINOv2-S, pointwise → statistics)	+1.00	+0.001	−0.011	−0.007	0.655 → 0.857
C9 → C10 (CT dose network, pointwise → statistics)	−0.09	+0.009	−0.011	−0.007	0.871 → 0.875
C5 → C6 (pointwise, DINOv2-S → DINOv2-B)	+0.16	−0.008	+0.004	+0.007	0.655 → 0.716

- Matching: statistics improve DINOv2-S on every metric.
- CT network: the gain from statistics is small.
- Size: DINOv2-B shows no benefit and is 1.3× slower.
- No significance claim. Tests follow with three seeds.

Screen versus trained outcome

preliminary

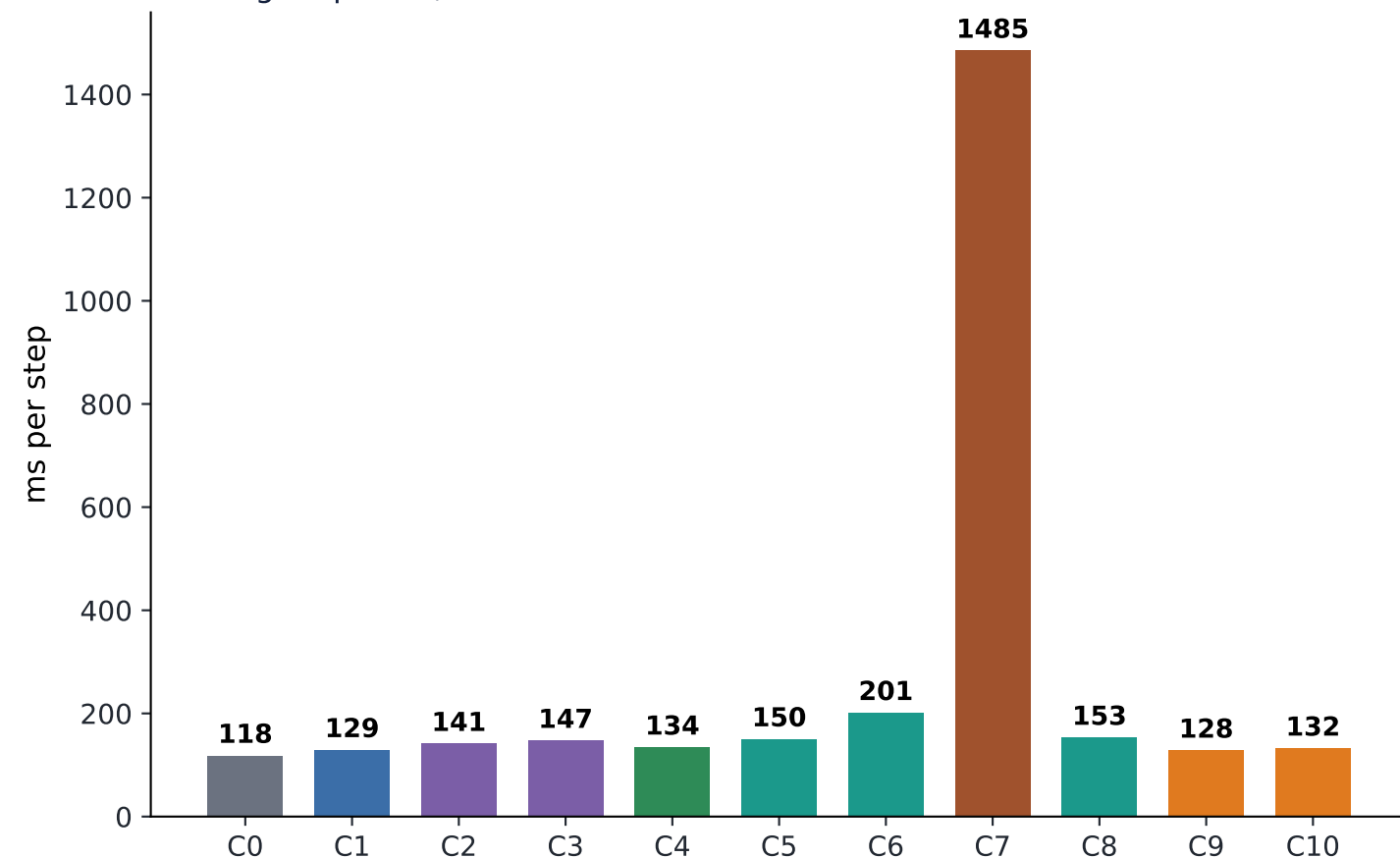
M: how well the distance orders the dose levels (1 = perfect). B: distance of a blurred image relative to the noisy 100k image; below 1, blur is cheap.

R_H: high-frequency content relative to the target (1 = same texture, lower = smoother output).

Screen signal	What it implied	What training gave (seed 42)	Verdict
Low M CT net 0.72	Distance barely tracks dose, so a weak quality signal	Worst DISTs of all perceptual losses (C9 0.100, C10 0.093)	consistent
B < 1 CT net 0.31	Blur is cheap, so the output should be smoothed (low R_H)	Opposite: R_H 0.87 and 0.88, the sharpest of all runs	contradicted
B > 1 all pretrained	Blur is penalised, so the output should keep its texture	Still smooth: R_H 0.58 to 0.86 (C7 MedSigLIP lowest)	contradicted
Lower B with statistics	Local statistics reward matching texture, not exact pixels	C8 (DINOv2-S, statistics) B 1.26, R_H 0.857 vs C5 (DINOv2-S, pointwise) B 1.75, R_H 0.655	consistent

- M held up: the CT net, the only low-M network, had the worst DISTs of the perceptual losses.
- B did not: its threshold of 1 failed in both directions. Among pretrained networks, lower B did go with higher R_H.
- The screen is a prediction, not a filter: all of C1 to C10 were trained regardless.

Training step time, A100 80 GB



Item	GPU hours
C0 run	0.155
C7 run	1.445
Seed 42, 11 runs	3.26
Used, Weeks 1 and 2	3.69
Projected, 3 seeds	10.2

- Budget is sufficient. No runs need cutting.
- C7 (MedSigLIP, pointwise) costs about 10× a C5 (DINOv2-S, pointwise) run.

Takeaways

Week 1	Pipeline verified. The earlier SSIM anomaly is not reproduced.
Week 2	Perceptual terms reduce LPIPS by 20 to 55% at a cost of 0.15 to 2 dB. C8 (DINOv2-S, statistics) is the best trade-off.
Matching	Statistics outperform pointwise for DINOv2-S. Larger and medical encoders add cost only.
Open	The CT dose network matches grain but not structure. Diagnosis comes first.
Status	One seed. Statistical tests follow in Week 3.

Appendix: full training-free screen (validation)

appendix

Network	Matching	M	Dose ratio 10k/1000k	B	Params
Pixel L1	-	1.000	5.74	0.886	0
VGG-19 relu3_2	pointwise	1.000	2.61	1.509	1.1 M
VGG-19 relu3_2	statistics	0.999	1.68	1.418	1.1 M
LPIPS-VGG	pointwise	0.998	2.43	1.483	14.7 M
DISTS	statistics	0.984	2.75	1.450	14.7 M
RadImageNet R50	pointwise	0.954	7.10	1.812	25.6 M
RadImageNet R50	statistics	0.933	3.60	1.396	25.6 M
DINOv2-S	pointwise	0.998	7.71	1.752	22.1 M
DINOv2-S	statistics	0.997	2.37	1.260	22.1 M
DINOv2-B	pointwise	0.997	6.27	1.669	86.6 M
DINOv2-B	statistics	0.994	2.31	1.250	86.6 M
DINOv2-L (screen only)	pointwise	0.996	6.89	1.874	304 M
DINOv2-L (screen only)	statistics	0.997	2.32	1.278	304 M
CLIP ViT-B/16 (screen only)	pointwise	0.996	2.73	1.571	85.8 M
CLIP ViT-B/16 (screen only)	statistics	0.990	1.75	1.419	85.8 M
MedSigLIP-448	pointwise	1.000	3.25	1.535	429 M
MedSigLIP-448	statistics	0.996	2.33	1.537	429 M
CT dose network	pointwise	0.724	4.93	0.308	1.2 M
CT dose network	statistics	0.712	2.69	0.754	1.2 M